

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (CL)

May 2012

CL 204 Engineering Geology

Date: 04.05.2012, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Draw neat figures wherever required. Also quote suitable examples.

SECTION - I

- Q - 1 What is seismology and seismic activity? Explain characteristics of various seismic waves. [07]
- Q - 2 (a) Define rock and mineral. Enlist various branches of geology. Write the scope of Geology in construction job. [05]
- (b) Write about denudation. Explain physical weathering on the rock mass. [05]
- (c) Define and explain briefly the following terms: [04]
- (i) Colour and luster in minerals (ii) Structures in minerals

OR

- Q - 2 (a) How sedimentary rocks are formed? Enlist mechanical structures developed in sedimentary rocks. [05]
- (b) Write short notes on: [05]
- (i) Weathering due to plants (ii) Carbonation process on rocks
- (c) Write note on uses of various geological materials as construction material. Give suitable examples. [04]
- Q - 3 (a) How igneous rocks are formed in nature? Enlist light and dark colour minerals. [05]
- (b) (i) Write a note on texture in sedimentary rocks. [03]
- (ii) Explain soil type based on sieve analysis. [02]
- (c) (i) Explain briefly metamorphism due to heat and pressure. [04]

OR

- Q - 3 (a) Write a note on various structures observed in igneous rocks. [05]
- (b) What is metamorphism? Write metamorphic effect on various rocks with suitable examples. [05]
- (c) Differentiate between magnitude and intensity of an earthquake. [04]

SECTION - II

- Q - 4 What is landslide? Write note on various causes of mass movement. [07]

- Q - 5 (a) What is fault and faulting in rock? Draw figures showing normal and reverse faults and define their parts. [05]
- (b) Draw a neat sketch of ideal folds and define the related terminologies. [05]
- (c) Give detailed classification of joints. [04]

OR

- Q - 5 (a) How fold and faults are associated with plate tectonic activity? [05]
- (b) Write a note on various unconformities. Also write their significance. [05]
- (c) How trend and rate of weathering affects reservoir for its life span? [04]
- Q - 6 (a) Geologically, how pore pressure and weight of dam has influence on site selection for dam? [05]
- (b) What geological investigations are to be carried out for selecting tunnel alignment? Add your note with tunnel in hard rock formation. [05]
- (c) Explain confined and unconfined aquifer. [04]

OR

- Q - 6 (a) How altitudes of beds have influence on site selection for a dam? Discuss various conditions. [05]
- (b) Discuss influence of ground water conditions in tunneling work. [05]
- (c) Discuss importance of geological maps in site selection of civil projects. [04]
